

VEMULA THUSHAR REDDY

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EDUCATION

Indraprastha Institute of Information Technology, Delhi

B.Tech. ECE

CGPA: 7.88

Dec 2020 - May 2024

Narayana Junior College Madhapur, Hyderabad

Telangana State Board of Intermediate Education

Percentage: 92.7%

2018 - 2020

SKILLS

Programming Languages Python, Java, SQL, MATLAB, R

Tools and Technologies Git, Github, MySQL Workbench, HTML, CSS, Matplotlib, Pandas, Numpy, OpenCV, Manim

Relevant Coursework Data Structures and Algorithms, Advanced Programming, Database Management And System, Computer Organisation, Convex Optimization, Human-Computer Interaction, Quantitative Biology, Robotics, Computer Architecture

WORK EXPERIENCE

Midas Research Associate

Aug 2022 - Dec 2023

- Engaged in the curation and development of datasets for the "IndicNLP" initiative in Telugu.
- Conducted analysis of datasets, integrating valuable website enhancements, including features such as BLEU score, word cloud, word frequency, and IAA score.

Undergraduate Student Researcher

May 23 - present

- Conducted research on collected datasets and collected nearly 40k protein fasta files.
- Utilized dbCAN annotation tool to identify CAZy families in genomes and generated the CAZy matrix using HMMER.
- Actively participated in research through careful data analysis and project management.

PROJECTS

Path Following Robot | [Github](#)

July 2023 - Dec 2023

- Developed a Python simulation using Pygame for rendering.
- Implemented dynamic visibility map and following field for guiding a virtual robot through mazes.
- Enabled real-time route updates in response to obstacles and changing visible surroundings.

Online Retail Store | [Github](#)

Jan 2023 - Apr 2023

- Developed a command line interface of a retail store with CRUD functionalities.
- Achieved a streamlined working CLI with Flask and MySQL for efficient data management.

Cache coherence protocols | [Github](#)

July 2023 - Dec 2023

- Developed a multi-core cache coherence system simulation.
- Implemented and analyzed a cache coherence protocol to ensure data consistency.
- Assessed system behavior, optimizing for performance and scalability.

Snakes & Ladders | [Github](#)

Jan 2023 - Apr 2023

- Developed an interface iteration of the popular Board Game - Snakes and Ladders using object-oriented and event-driven programming.
- Utilised the Serializable Interface for saving games progress.

ACHIEVEMENTS

- Captained** the IIIT Delhi Volleyball team and making significant contributions to the Team's performance in campus tournaments.
 - Antaranga - IIITD's inter-branch sports competition (2022)
 - Astra - IIITD's sports fest (2023)
 - Utkarsh - NSUT's sports fest (2021)
 - Rann-Neeti - IIT Mandi's sports fest(2022)

Date: 11 Dec 2023